COMSATS University Islamabad, Lahore Campus

☑ Midterm Examination – FALL 2022

Course Title:	Design and Analysis of Algorithms				**Course Code:**	CSC301	**Credit Hours:** 3(3,0
Course Instructor:	Dr. Hasan Jamal				**Programme Name:**	BS Computer Science	
Semester;	5th	**Batch:**	FA20-BCS	**Section:**	A, B	**Date:**	18/11/2022
Time Allowed:	90 Minutes				**Maximum Marks:**		50
Student's Name:					**Reg. No.**		

Important Instructions / Guidelines:

- Be precise and to the point while answering any question
- Show all your work, as partial credits will be given. You will be graded not only on the correctness of your answer, but also on the clarity with which you express it. Please be neat.
- **Good luck!**

Question 1: **CLO: <1>; Bloom Taxonomy Level: <Understanding>** **[Marks:**

A mystery problem can be solved using two algorithms that have the following time complexity:

$$T_A = 2n^2 - 2n + 8$$
$$T_B = 2n^2 + 6n - 12$$

Since both algorithms are $\Theta(n^2)$, we cannot use asymptotic notations to determine which algorithm is *better. How will we determine which algorithm is better? Also, determine the value of* n_0.

Question 2: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks:**

Prove or disprove: $\frac{1}{3}n^2 - \frac{1}{3}n - 3 \in \Theta(n^2)$

Question 3: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks:**

For the following code snippet, provide a line-by-line analysis and construct function T(n) that give the runtime of this code snippet as a function of "n". Also determine the Big-Oh of this code snippet.

```
x = n;
while (x > 0) {
        if ( x < i)
            sum += foo( x ) * foo( x-1);
        else
            sum += bar( x );
        x = x - 1
}

bar (a) {
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < i; ++j)
                sum += a*(i + j);
        return sum;
}

foo (a) {
        for (int i = 0; i < n; ++i)
                sum += a * i;
        return sum;
}
```

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Question 4: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 10]**

In not more than five lines each, provide the answers to the following questions. Be reasonable in your assumptions. Justify your answer.

a) You are given an unsorted list of 10,000 items with repeating numbers ranging from 0 to 10,000. Which sorting algorithm will you choose to sort this data set? Justify your answer.

b) You have a uniformly distributed set of unsorted integers of 50,000 numbers, within the range 0 to 100,000. Which algorithm will you choose to sort this data set? Justify your answer.

c) You have a list of share prices of 500 stocks. To sort this list, you need a stable sorting algorithm with an absolute guarantee of O(n lg n) performance in the worst case. Which algorithm will you choose? Justify your answer.

d) Suppose you have found a piece of code from "Stackoverflow" that takes a list of sorted numbers as input and perform some operations on the sorted list. However, unfortunately, there is a bug in the code that randomly shuffles a few numbers. To save time, instead of fixing the bug, you decide to sort this list again. Which algorithm will you choose to sort this list? Justify your answer.

Question 5: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 10]**

Solve the following recurrence using the "Recursion Tree Method"

$$T(n) = 3\,T\left(\frac{n}{3}\right) + 1$$

Question 6: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 5]**

Solve the following recurrence using the "Master Theorem Method"

$$T(n) = T\left(\frac{n}{4}\right) + 1$$

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